UDACITY
Part of Accenture

Calculation of Basketball Scoring Probabilities

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Agenda

Results of basketball scoring calculations

- Overall Shooting Statistics
- Probabilities
- Conditional Probabilities - Future
- Conditional Probabilities - Retrospective

Overall Shooting Statistics

Total shots analyzed : 1,434

P(Make) : 0.4902 (49.02%)

P(Miss) : 0.5098 (50.98%)

P(3-pointer) : 0.5481 (54.81%)

P(2-pointer) : 0.4519 (45.19%)

Probabilities

- P(exactly 3 made of next 4 shots) : 0.2402 (24.02%)
- P(exactly 4 of next 5 are 3-pointers) : 0.2039 (20.39%)
- Assumptions: fixed n, binary outcomes, independent trials, constant p.
- Independence and constant p hold only approximately for shot data.

Conditional Probabilities - Future

P(Made | 3-pointer) : 0.4186 (41.86%)
P(Lead | 3-pointer) : 0.4936 (49.36%)
P(Made | 2-pointer) : 0.5772 (57.72%)
P(Lead | 2-pointer) : 0.5154 (51.54%)

Supporting quantities:

P(Made) : 0.4902 (49.02%)
P(3) : 0.5481 (54.81%)
P(2) : 0.4519 (45.19%)
P(Lead) : 0.5035 (50.35%)
P(3 | Lead) : 0.5374 (53.74%)
P(2 | Lead) : 0.4626 (46.26%)
P(Made | 3) x P(3) : 0.2294 (22.94%)
P(Made | 2) x P(2) : 0.2608 (26.08%)

Conditional Probabilities - Retrospective

P(3-pointer | Made) : 0.4680 (46.80%)
P(2-pointer | Made) : 0.5320 (53.20%)
Sanity check (sum = 1.0) : 1.000000